

Python Automation for Engineers

Course Duration: 3 Days (21 Hours)

Course Description

This practical, hands-on course equips engineers with Python automation skills to eliminate repetitive tasks—log analysis, test result processing, report generation, and data pipeline management. Built on a Problem → Concept → Implementation → Practice model, participants leave with working automation scripts they can deploy immediately.

Learning Outcomes

- Write Python scripts to automate real engineering tasks from day one
- Parse, transform, and summarize log files and structured data (CSV, JSON)
- Build reusable, maintainable automation libraries for team use
- Integrate error handling and scheduling into automation pipelines
- Perform data analysis to detect anomalies and generate visual summaries
- Complete a capstone automation project aligned to your own workflow

Prerequisites

- Basic understanding of **programming or scripting concepts** in any language (such as C, C++, Java, MATLAB, Bash, or similar)
- Familiarity with **engineering or software development workflows**
- Basic experience working with **files, logs, or structured data (e.g., CSV, JSON)**
- Comfort using a **computer development environment or command-line tools**

Target Audience

- Software Engineers
- Test Engineers
- System Engineers
- DevOps Engineers
- Data Engineers
- Technical Analysts

Course Outline

DAY 1

Python Fundamentals for Engineering Automation (7 Hours)

MODULE 1 Introduction to Python for Engineers ⌚ 1.5 hrs

- Why automation matters for engineering teams
- Identifying repetitive tasks ripe for automation
- Python as an engineering automation tool
- Setting up the Python development environment
- Running and structuring Python scripts

Getting Started Lab

- Install and configure Python environment
- Run and modify starter scripts
- Write your first engineering automation script

MODULE 2 Python Programming Essentials ⌚ 2.5 hrs

- Variables, data types, and operators
- Lists, tuples, and dictionaries
- Conditional logic and branching
- Loops, iterations, and comprehensions
- Writing reusable, parameterized functions

Data Scripting Lab

- Process and filter engineering data files
- Automate repetitive calculations
- Build a multi-function utility script

MODULE 3 Working with Files & Data ⌚ 3 hrs

- Reading and writing text files
- Parsing structured and unstructured log files
- Automating file operations (rename, move, archive)
- Extracting key fields with string parsing & regex
- Generating automated file summaries

Log Processing Lab

- Read and parse real system log samples
- Extract error counts, timestamps, and patterns

- Generate an automated plain-text summary report

DAY 2 Automating Engineering Workflows (7 Hours)

MODULE 4 Advanced Data Processing ⌚ 2.5 hrs

- Reading and writing CSV datasets with Python
- Parsing and generating JSON data
- Data transformation and normalization techniques
- Filtering, grouping, and summarizing datasets
- Converting log output into structured formats

Structured Data Lab

- Convert raw log data to clean CSV/JSON
- Aggregate metrics from engineering datasets
- Produce a structured engineering data report

MODULE 5 Automation Techniques & Error Handling ⌚ 2.5 hrs

- Designing reliable automation workflows
- Batch processing multiple files automatically
- Script scheduling concepts (cron, task scheduler)
- Robust error handling and logging in scripts
- Generating automated alerts on threshold breaches

Workflow Automation Lab

- Build a batch file processor with error handling
- Monitor a log directory for new entries
- Trigger automated email/file alerts on anomalies

MODULE 6 Data Analysis for Engineering Insights ⌚ 2 hrs

- Basic statistical analysis with Python
- Identifying patterns and trends in log data
- Anomaly and outlier detection techniques
- Using Python plotting libraries for visual summaries
- Applying analysis to troubleshooting workflows

Analysis & Visualization Lab

- Analyze a multi-day engineering log dataset

- Detect anomalies and performance degradations
- Generate charts and a visual summary report

DAY 3 Practical Automation Projects (7 Hours)

MODULE 7 Building Production-Ready Automation Scripts ⌚ 2 hrs

- Principles of clean, maintainable script design
- Structuring reusable modules and libraries
- Configuration management (config files, env vars)
- Version control basics for automation scripts
- Documenting and packaging scripts for team use

Script Architecture Lab

- Refactor a working script into a reusable module
- Add configuration file support
- Implement logging and structured output

MODULE 8 Engineering Automation Case Studies ⌚ 2 hrs

- Automation patterns in software testing pipelines
- DevOps workflow automation (CI/CD integration)
- System monitoring and alerting automation
- Test result aggregation and reporting automation
- Lessons from real-world automation implementations

Case Study Workshop

- Analyze 2–3 industry automation scenarios
- Identify automation opportunities and risks
- Design an automation solution for each case

MODULE 9 CAPSTONE PROJECT Mini Automation Project (3 Hours)

Participants independently build a complete automation solution for a realistic engineering problem, demonstrating all skills acquired across the 3 days.

Example Project Tracks:

- **Automated Log Analyzer**
Parse multi-source logs, detect errors, output summary dashboard
- **Test Result Reporter**
Aggregate test outputs, compute pass rates, generate HTML report

- **Data Processing Pipeline**

Ingest raw CSVs, transform, validate, and export clean datasets